

$$\frac{\partial u}{\partial t} = f(u) \iff \frac{\partial u}{\partial t} = v \frac{\partial u}{\partial x^2}$$

u is a vector.

Recall Euler

$$u^{n+1} = u^n + \Delta t f(u^n) \quad \text{Error is } o(\Delta t)$$

Extend to Runge Kutta family

Let $u^{n+1} = u^n + \Delta t f(u^*)$, u^* is an approximation.

Take u^* to be Euler by a smaller time step

$$u^* = u^n + \frac{\Delta t}{2} f(u^n)$$

$$\Leftrightarrow u^{n+1/2} = u^n + \frac{\Delta t}{2} f(u^{n+1/2})$$

This is a predictor-corrector method

or a mid point method.

Another view:

$$\text{Let } k_1 = f(u^n)$$

$$k_2 = f(u^n + \frac{\Delta t}{2} k_1)$$

$$u^{n+1} = u^n + \Delta t k_2$$

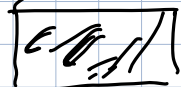
This is called 2nd order Runge Kutta or RK2

Runge-Kutta-Heun scheme.

$$k_1 = f(u^n)$$

$$k_2 = f(u^n + \Delta t k_1)$$

$$u^{n+1} = u^n + \frac{1}{2} (k_1 + k_2)$$

$$f(u^n) \quad f(u^n + \Delta t k_1)$$


Runge-Kutta-Ralston:

$$k_1 = f(u^n)$$

$$k_2 = u^n + \frac{2}{3} \Delta t k_1 = u^n + \frac{2}{3} \Delta t f(u^n)$$

$$u^{n+1} = u^n + \Delta t \left[\frac{1}{4} k_1 + \frac{3}{4} k_2 \right]$$

Idea:

$$\begin{aligned}
 u(t_{n+1}) &= u(t_n) + \int_{t_n}^{t_{n+1}} f(u^n) dt \\
 &= u(t_n) + f(u^n) (t_{n+1} - t_n) \\
 &= u(t_n) + \Delta t f(u^n)
 \end{aligned}$$

Above mentioned schemes are $O(\Delta t^2)$

4th order Runge-Kutta $\Rightarrow$ RK4

$$k_1 = f(u^n)$$

$$k_2 = f(u^n + \frac{\Delta t}{2} k_1)$$

$$k_3 = f(u^n + \frac{\Delta t}{2} k_2)$$

$$k_4 = f(u^n + \Delta t k_3)$$

$$u^{n+1} = u_n + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

Lead error is $O(\Delta t^4)$

Amplification factor of RK4

$$G(z) = \frac{u^{n+1}}{u^n} \quad \text{or} \quad R(z) = \frac{u^{n+1}}{u^n}$$

Derivation based on $\frac{\partial u}{\partial t} = \lambda u = f(u)$

$$k_1 = f(u^n) = \lambda u^n$$

$$\begin{aligned} k_2 &= f(u^n + \frac{\Delta t}{2} k_1) = \lambda \left(u^n + \frac{\Delta t}{2} \lambda u^n \right) \\ &= \lambda u^n \left(1 + \frac{\lambda \Delta t}{2} \right) \\ &= \lambda u^n \left(1 + z/2 \right), \quad z = \lambda \Delta t \end{aligned}$$

$$\begin{aligned} k_3 &= f(u^n + \frac{\Delta t}{2} k_2) \\ &= \lambda \left(u^n + \frac{\Delta t}{2} \lambda u^n (1 + z/2) \right) \\ &= \lambda u^n \left(1 + \frac{z}{2} (1 + z/2) \right) \\ &= \lambda u^n \left(1 + z/2 + z^2/4 \right) \end{aligned}$$

$$\begin{aligned} k_4 &= f(u^n + \Delta t k_3) \\ &= \lambda \left(u^n + \Delta t \left[\lambda u^n (1 + z/2 + z^2/4) \right] \right) \\ &= \lambda u^n \left(1 + z (1 + z/2 + z^2/4) \right) \\ &= \lambda u^n \left(1 + z + z^2/2 + z^3/4 \right) \end{aligned}$$

$$\begin{aligned} u^{n+1} &= u^n + \frac{\Delta t}{6} \left[\underbrace{\lambda u^n}_{(k_1)} + 2 \underbrace{\lambda u^n (1 + z/2)}_{(2k_2)} + 2 \underbrace{\lambda u^n (1 + z/2 + z^2/4)}_{(2k_3)} \right. \\ &\quad \left. + \lambda u^n (1 + z + z^2/2 + z^3/4) \right] \end{aligned}$$

$$\begin{aligned} &= u^n + \frac{\Delta t}{6} (\lambda u^n) \left[1 + 2(1 + z/2) + 2(1 + z/2 + z^2/4) \right. \\ &\quad \left. + (1 + z + z^2/2 + z^3/4) \right] \end{aligned}$$

$$\frac{u^{n+1}}{u^n} = 1 + \frac{z}{6} \left(6 + 3z + z^2 + \frac{z^3}{4} \right)$$

$$G(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}$$

Remark: A-stability requires $|G(z)| < 1$

Ex: $\frac{\partial u}{\partial t} = \lambda u$

Show $u(t_{n+1}) = e^{\lambda \Delta t} u(t_n)$

Multi step methods.

① two step: $\frac{\partial u}{\partial t} \approx \frac{u^{n+1} - u^{n-1}}{2\Delta t}$

Then $\frac{\partial u}{\partial t} = f(u) \Rightarrow \frac{u^{n+1} - u^{n-1}}{2\Delta t} = f(u^n)$

$$u^{n+1} = u^{n-1} + 2\Delta t f(u^n)$$

lead error $O(\Delta t^2)$

Starting problem

At $n=0$, $u^1 = u^0 + 2\Delta t f(u^0)$

↑
not available.

At $n=1$, $u^2 = u^0 + 2\Delta t f(u^1)$

To solve starting problem

$$u^1 = u^0 + \Delta t f(u^0)$$

Euler method

The above scheme is called Leap-Frog method

(2) BDF - scheme

$$\frac{\partial u}{\partial t} = \frac{3u^{n+1} - 4u^n + u^{n-1}}{2\Delta t} = \frac{\partial u}{\partial t}(t+\Delta t) + \frac{1}{12} \Delta t^2 \frac{\partial^2 u}{\partial t^2}(t+\Delta t)$$

$$\approx f(u^{n+1})$$

(need to derive)
H.W.

Remark: Multistage and multi step methods are useful for non-linear problems.

Ex:

$$\frac{\partial \vec{u}}{\partial t} = \begin{cases} \frac{\partial u}{\partial t} = -k_1 uv + k_2 w, \\ \frac{\partial v}{\partial t} = -k_1 uv + k_2 w, \\ \frac{\partial w}{\partial t} = k_1 uv - k_2 w \end{cases} \begin{cases} K_1, K_2 \text{ constants} \\ f(\vec{u}) \end{cases}$$

let $\vec{u}(t) = \bar{u} + u'(t)$, $\vec{u}(t) = \begin{bmatrix} u(t) \\ v(t) \\ w(t) \end{bmatrix}$

Then $f(\vec{u}(t)) = f(\bar{u} + u'(t))$
 $= f(\bar{u}) + \nabla f \cdot u' + \text{Higher order terms}$

$$\frac{\partial u'}{\partial t} = \frac{\partial}{\partial t}(\vec{u} - \bar{u}) = \frac{\partial \vec{u}}{\partial t} - \frac{\partial \bar{u}}{\partial t}$$

$$= f(\vec{u}) - 0, \text{ if } \bar{u} \text{ does not depend on } t$$

$$= f(\bar{u}) + \nabla f \cdot u'$$

$$= \frac{\partial \bar{u}}{\partial t} + \nabla f \cdot u'$$

$$= \nabla f \cdot u'$$

$$\frac{\partial u'}{\partial t} = \nabla f \cdot u'$$

$\Downarrow$

$$\frac{\partial u'}{\partial t} = \Lambda u' \quad \neq \quad \Lambda u'$$

